Task 5 – Asymptotic analysis

Linear search: best $\Omega(1)$, worst $O(n)$, average $\Theta(n)$. Recursive binary search: best $\Omega(1)$, worst $O(\log n)$, and average $\Theta(\log n)$, provided input is sorted. Hash lookup: expected $\Theta(1)$, worst $O(n)$, plus $\Theta(n)$ expected index construction. As n increases from 1,000 to 1,000,000, linear search may require roughly 1,000 times as many comparisons in a comparable worst-case query. Binary search adds only about $\log_2(1{,}000{,}000) \approx 20$ levels. Hash lookup is expected to remain near one indexed lookup per request, but consumes extra memory and relies on good hashing.